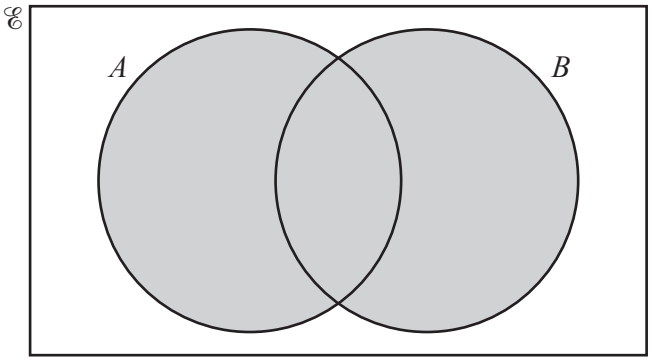
0580_s22_qp_33_Q2

2 **(a)**



Use set notation to describe the shaded region.

..... [1]

(b) $\mathscr{E} = \{x : x \text{ is a natural number} \leqslant 16\}$

(i) Write down all the square numbers in the universal set, $\mathscr{E}$.

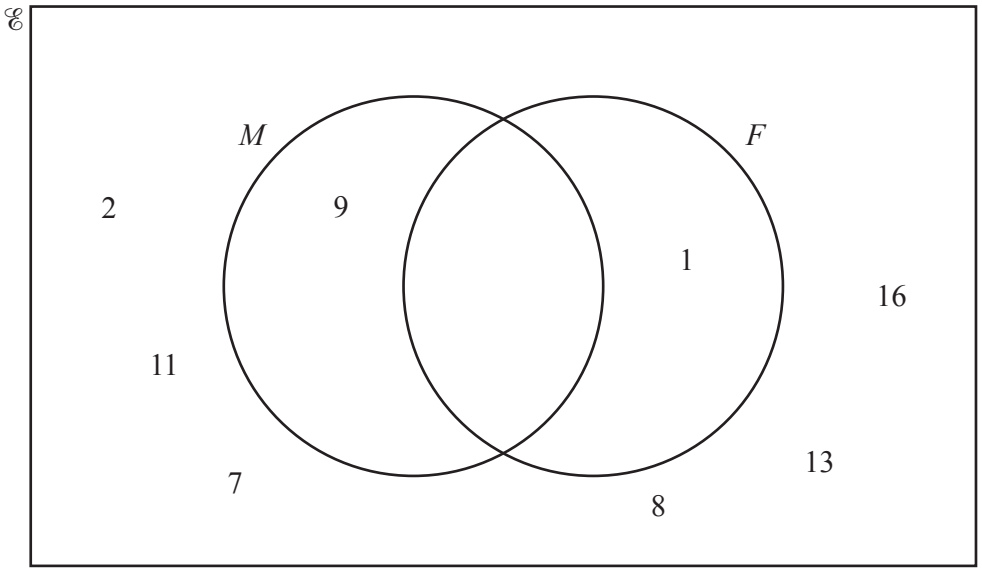
..... [2]

(ii) Write down the six prime numbers in the universal set, $\mathscr{E}$.

....., ..., ..., ..., ..., ... [2]

(iii) $M = \{x : x \text{ is a multiple of } 3\}$
$F = \{x : x \text{ is a factor of } 15\}$

(a) Complete the Venn diagram to show the elements of these sets.



[2]

(b) Write down all the odd numbers that are not in set M and not in set F.

..... [1]

(c) Find $n(M \cap F)$.

..... [1]

(d) A number is chosen at random from the universal set, $\mathscr{E}$.

Find the probability that this number is in set F.

..... [1]